

Relieve track hotspots

When track overflow appears, detailed routers rip up contributing nets and reroute around congested tracks

The idea

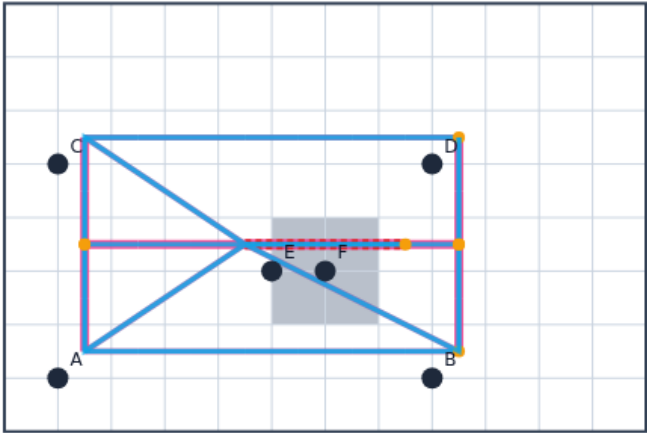
- Score each net by track overflow on its segments
- Subtract its route from usage
- Run `astar_route` between its pins with the remaining usage map
- Add the new segment track usage back
- Total overflow should not rise; ideally it drops versus the pre-rip state

Sequential first

RIP-UP DETAILED

Sequential first

Step 1 / 5 · seq · module03-03-ripup-detailed



Explanation

Route layered; tracks exceed Cap.

- Overflow appears

total=3

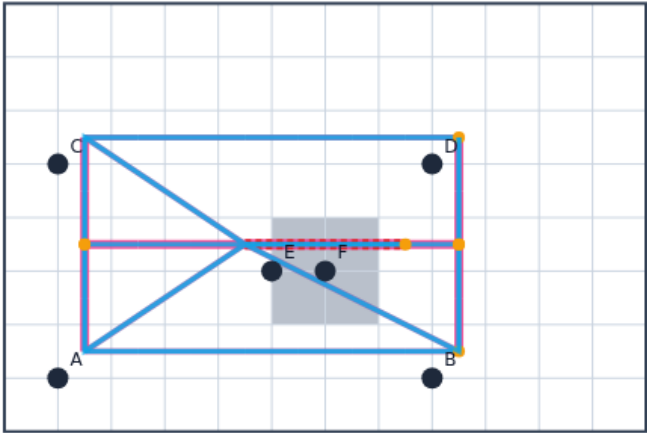
Sequential first

Pick hot net

RIP-UP DETAILED

Pick hot net

Step 2 / 5 · pick · module03-03-ripup-detailed



Explanation

Find net with worst overflow contribution.

- One net rip

Toy order

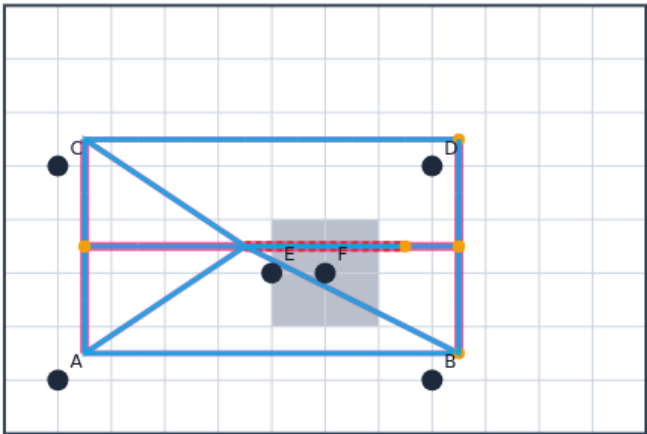
Pick hot net

Rip segments

RIP-UP DETAILED

Rip segments

Step 3 / 5 · rip · module03-03-ripup-detailed



Explanation

Subtract ripped net from track usage.

- Ready to reroute

Before reroute

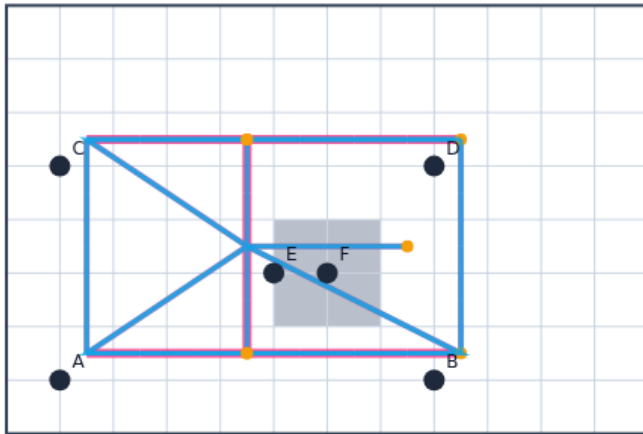
Rip segments

A* reroute

RIP-UP DETAILED

A* reroute

Step 4 / 5 · astar-r · module03-03-ripup-detailed



Explanation

Reroute ripped net with A* on updated usage.

- May improve

`GOLDENS.ripupImproves`

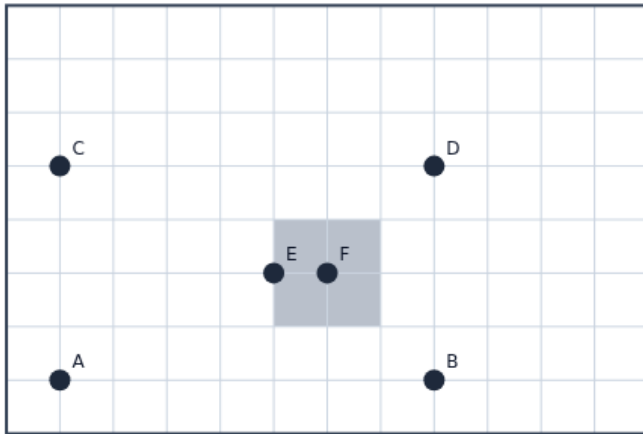
A* reroute

Iterate

RIP-UP DETAILED

Iterate

Step 5 / 5 · loop · module03-03-ripup-detailed



Explanation

Real detailed routers loop route → DRC → rip until clean.

- Course wrap

```
learn_drc
```

Iterate

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Rip-up detailed reroute

INTERACTIVE TOOL

Rip-up detailed reroute

Rip the worst overflowing net and reroute with A*.

Your job: Route layered, note overflow, click Rip-up detailed. Golden: rip-up can reduce total on hot tracks.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Route layered first

Pick a challenge and click **Start**. Move cells, route, then **Check**.

StartShow hintCheckNext challengeStop checkingIdle

A B C D E F←→↑↓Route LeeRoute layeredRoute A*

Rip-up detailed

Graph

Reference / metrics

selected: A @ grid (1,1)
routes: (none - click Route)
overflow total: 0.00 · max: 0.00 · count: 0
track capacity: 2 · mode: astar
HPWL: 34.0

Browser lab starter

learn_routing — 03 ripup detailed

Implement track

- Implement ``ripup_detailed(routes, usage, cap, terminals_map, nets, nx, ny, blocks)``
- Assert total overflow after is less than or equal to before on `tiny_dr` sequential L seed

Pitfalls

- Ripping a net but leaving ghost usage on its old tracks
- Rerouting with pattern L through the same hot track
- Picking the wrong net to rip, use overflow contribution not HPWL

Your turn

- Clear rip-up challenges
- Next: tie it together with full sequential detailed routing